

SACHEON AB, Korea, Republic of

WMO: 471610

Lat: 35.089N

Lon: 128.070E

Elev: 8

StdP: 101.23

Time Zone: 9.00 (E09)

Period: 82-19

WBAN: 43240

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.0	(c) -8.8	(d) -16.0	(e) 0.9	(f) -2.6	(g) -14.1	(h) 1.1	(i) -3.4	(j) 7.5	(k) 1.3	(l) 6.4	(m) 2.0	(n) 0.8	(o) 60	(p) 0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.2	(c) 33.1	(d) 26.5	(e) 31.9	(f) 25.8	(g) 30.2	(h) 24.9	(i) 27.1	(j) 31.5	(k) 26.5	(l) 30.5	(m) 26.0	(n) 29.4	(o) 3.4	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	29.8	25.2	20.4	28.8	24.9	20.0	28.5	85.6	31.8	83.0	30.7	80.5	29.6	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.0	(b) 6.0	(c) 5.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	-0.1	2.0	6.8	12.8	17.7	21.7	25.4	26.5	21.4	15.4	9.2	3.2
	DBStd	9.36	2.70	3.79	3.10	3.31	2.32	1.99	2.47	2.02	2.87	3.17	3.37	3.30
	HDD10.0	923	313	226	105	11	0	0	0	0	3	53	213	
	HDD18.3	2446	572	458	358	168	40	2	0	0	6	98	274	471
	CDD10.0	2219	0	1	6	96	237	351	478	511	341	170	29	1
	CDD18.3	700	0	0	0	2	19	103	220	252	97	7	0	0
	CDH23.3	5766	0	0	0	14	177	661	1853	2343	646	73	0	0
	CDH26.7	1818	0	0	0	1	13	143	628	894	137	2	0	0
(14) Wind	WSAvg	1.9	1.5	1.8	2.2	2.4	2.2	2.0	2.3	1.9	1.6	1.6	1.6	1.6
(15) Precipitation	PrecAvg	1583	28	52	84	139	152	191	312	306	180	61	52	28
	PrecMax	2285	60	138	166	296	294	367	688	757	448	195	196	101
	PrecMin	872	0	0	14	31	53	49	58	79	15	0	0	0
	PrecStd	360	16	36	41	65	62	105	162	182	137	48	47	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	15.9	18.9	25.0	28.1	31.1	35.2	35.0	31.2	26.9	21.2	16.2
		MCWB	6.9	11.0	11.5	17.4	19.2	22.0	26.4	27.2	24.7	18.6	16.3	11.7
	2%	DB	9.7	12.8	16.9	23.1	26.8	29.8	33.1	33.5	29.8	25.1	19.8	14.0
		MCWB	5.3	8.8	10.8	15.7	18.5	21.5	26.5	26.6	23.7	18.3	14.1	9.6
	5%	DB	7.9	10.5	15.1	21.3	25.2	28.2	31.2	32.2	28.1	23.9	18.0	12.1
		MCWB	3.9	6.3	9.6	14.7	17.8	21.1	26.0	26.0	22.5	17.6	12.7	7.7
	10%	DB	6.1	9.0	13.8	19.9	24.0	27.0	30.0	31.0	27.0	22.1	16.2	10.1
		MCWB	2.4	5.8	8.7	14.4	16.9	21.0	25.5	25.8	21.9	16.5	12.0	6.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	13.2	14.2	19.7	21.9	24.7	27.7	28.2	26.2	21.9	18.3	12.6
		MCDB	10.4	14.7	16.5	21.9	25.3	27.7	32.3	33.1	29.4	24.6	20.0	15.8
	2%	WB	5.7	9.1	12.1	17.9	20.6	23.7	27.0	27.2	25.3	19.9	15.4	10.3
		MCDB	8.6	10.5	15.2	21.1	23.6	26.1	31.5	31.8	28.3	22.9	18.1	13.4
	5%	WB	4.3	7.2	10.6	16.4	19.6	22.8	26.5	26.7	24.4	18.6	14.1	8.5
		MCDB	6.8	9.9	13.6	19.5	23.0	25.9	30.4	30.8	27.1	22.0	16.7	11.0
	10%	WB	3.1	5.8	9.3	15.1	18.5	22.1	25.9	26.2	23.0	17.4	12.6	6.9
		MCDB	6.0	8.3	12.4	18.9	22.2	25.7	29.3	29.9	25.7	21.1	15.8	9.5
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	11.9	12.4	12.4	11.5	9.6	7.2	8.2	10.0	13.0	13.1	13.5
		MCDBR	14.0	13.8	14.6	14.3	14.1	12.9	9.7	10.2	10.8	14.3	14.6	15.5
	5% WB	MCWBR	10.2	9.6	9.2	7.9	7.3	6.1	4.4	4.5	5.4	8.0	9.4	11.1
		MCWBR	12.6	11.5	12.3	11.0	10.6	8.9	8.4	8.7	8.1	11.6	12.1	14.8
(40) Clear-Sky Solar Irradiance	taub	0.427	0.477	0.557	0.585	0.599	0.651	0.612	0.568	0.507	0.471	0.443	0.421	
		taud	2.010	1.923	1.759	1.748	1.763	1.724	1.867	1.973	2.084	2.059	2.046	2.041
	Ebn at Noon	734	742	717	725	722	683	707	730	756	742	712	710	
		Edh at Noon	136	166	213	226	225	234	201	178	152	143	130	125
(44) All-Sky Solar Radiation	RadAvg	2.70	3.39	4.39	5.02	5.51	4.86	4.50	4.59	4.01	3.69	2.77	2.44	
	RadStd	0.20	0.31	0.38	0.40	0.58	0.44	0.65	0.59	0.43	0.34	0.30	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61	N/A

Nomenclature: See separate page